

全机型数据维度config说明

上海智元G1

代码块

```
1  A2D_CONFIG = {
2      "robot_type": "Genie1",
3      "arm_dim": 14,          # 左臂7 + 右臂7
4      "effector_dim": 2,      # 左夹爪 + 右夹爪
5      "cameras": ["head", "hand_left", "hand_right"],
6      "target_image_size": (640, 480), # (width, height) - 统一resize目标尺寸: 宽
640、高480
7      "motor_names": [
8          "left_arm_joint1",
9          "left_arm_joint2",
10         "left_arm_joint3",
11         "left_arm_joint4",
12         "left_arm_joint5",
13         "left_arm_joint6",
14         "left_arm_joint7",
15         "right_arm_joint1",
16         "right_arm_joint2",
17         "right_arm_joint3",
18         "right_arm_joint4",
19         "right_arm_joint5",
20         "right_arm_joint6",
21         "right_arm_joint7",
22         "left_gripper_joint1",
23         "right_gripper_joint1",
24     ],
25 }
26
27 STATE_DIM = A2D_CONFIG["arm_dim"] + A2D_CONFIG["effector_dim"] # 16
28 ACTION_DIM = STATE_DIM
```

郑州智元G1

代码块

```
1  A2D_CONFIG = {
2      "robot_type": "zhengzhou_zhiyuan_G1",
```

```

3     "arm_dim": 14,                # 左臂7 + 右臂7
4     "effector_dim": 2,           # 左夹爪 + 右夹爪
5     "cameras": ["head", "hand_left", "hand_right"],
6     "target_image_size": (640, 480), # (width, height) - 统一resize目标尺寸: 宽
640、高480
7     "motor_names": [
8         "left_arm_joint1",
9         "left_arm_joint2",
10        "left_arm_joint3",
11        "left_arm_joint4",
12        "left_arm_joint5",
13        "left_arm_joint6",
14        "left_arm_joint7",
15        "right_arm_joint1",
16        "right_arm_joint2",
17        "right_arm_joint3",
18        "right_arm_joint4",
19        "right_arm_joint5",
20        "right_arm_joint6",
21        "right_arm_joint7",
22        "left_gripper_joint1",
23        "right_gripper_joint1",
24    ],
25 }
26
27 STATE_DIM = A2D_CONFIG["arm_dim"] + A2D_CONFIG["effector_dim"] # 16
28 ACTION_DIM = STATE_DIM

```

智元A2

代码块

```

1  A2_CONFIG = {
2      "robot_type": "ZhiYuanA2",
3      "arm_dim": 14,                # 左臂7 + 右臂7
4      "finger_dim": 12,            # 左手6 + 右手6
5      "head_dim": 2,               # 头部2个关节
6      "cameras": ["head_front", "chest_left", "chest_right"],
7      "target_image_size": (640, 480), # (width, height) - 统一resize目标尺寸: 宽
640、高480
8      "motor_names": [
9          "idx13_left_arm_joint1",
10         "idx14_left_arm_joint2",
11         "idx15_left_arm_joint3",
12         "idx16_left_arm_joint4",
13         "idx17_left_arm_joint5",

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```

14         "idx18_left_arm_joint6",
15         "idx19_left_arm_joint7",
16         "idx20_right_arm_joint1",
17         "idx21_right_arm_joint2",
18         "idx22_right_arm_joint3",
19         "idx23_right_arm_joint4",
20         "idx24_right_arm_joint5",
21         "idx25_right_arm_joint6",
22         "idx26_right_arm_joint7",
23         "left_thumb_0",
24         "left_thumb_1",
25         "left_index",
26         "left_middle",
27         "left_ring",
28         "left_pinky",
29         "right_thumb_0",
30         "right_thumb_1",
31         "right_index",
32         "right_middle",
33         "right_ring",
34         "right_pinky",
35         "idx27_head_joint1",
36         "idx28_head_joint2"
37     ]
38 }
39
40 STATE_DIM = A2_CONFIG["arm_dim"] + A2_CONFIG["finger_dim"] +
A2_CONFIG["head_dim"] # 28
41 ACTION_DIM = STATE_DIM

```

方舟无限arx

代码块

```

1  ARX_LOONG_FPS = 30
2  ARX_LOONG_ROBOT_TYPE = "arx_loong"
3
4  # Image dimensions from actual data inspection
5  IMAGE_HEIGHT = 480
6  IMAGE_WIDTH = 640
7  IMAGE_CHANNELS = 3
8
9  # Joint axes names for 14-dim combined features:
10 # left_arm(6) + left_gripper(1) + right_arm(6) + right_gripper(1)
11 JOINT_AXES_NAMES = [
12     # left arm (6)

```

```

13     "arm_master_l_joint1",
14     "arm_master_l_joint2",
15     "arm_master_l_joint3",
16     "arm_master_l_joint4",
17     "arm_master_l_joint5",
18     "arm_master_l_joint6",
19     # left gripper (1)
20     "arm_master_l_joint7",
21     # right arm (6)
22     "arm_master_r_joint1",
23     "arm_master_r_joint2",
24     "arm_master_r_joint3",
25     "arm_master_r_joint4",
26     "arm_master_r_joint5",
27     "arm_master_r_joint6",
28     # right gripper (1)
29     "arm_master_r_joint7",
30 ]
31
32 # End-effector axes names (flattened):
33 # left_pos(3) + left_euler(3) + left_gripper(1) +
34 # right_pos(3) + right_euler(3) + right_gripper(1) = 14
35 END_EFFECTOR_AXES_NAMES = [
36     "left_pos_x", "left_pos_y", "left_pos_z",
37     "left_roll", "left_pitch", "left_yaw",
38     "left_gripper",
39     "right_pos_x", "right_pos_y", "right_pos_z",
40     "right_roll", "right_pitch", "right_yaw",
41     "right_gripper",
42 ]
43
44 # Define LeRobot Dataset features
45 ARX_LOONG_FEATURES = {
46     # Combined state position (left_arm 6 + left_gripper 1 + right_arm 6 +
47     # right_gripper 1 = 14)
48     "observation.state": {
49         "dtype": "float32",
50         "shape": (14,),
51         "names": {"axes": JOINT_AXES_NAMES},
52     },
53     # Combined state velocity
54     "observation.velocity": {
55         "dtype": "float32",
56         "shape": (14,),
57         "names": {"axes": JOINT_AXES_NAMES},
58     },
59     # Combined state effort

```

```

59     "observation.effort": {
60         "dtype": "float32",
61         "shape": (14,),
62         "names": {"axes": JOINT_AXES_NAMES},
63     },
64     # End-effector poses (position + euler angles + gripper, flattened)
65     "observation.end": {
66         "dtype": "float32",
67         "shape": (14,), # left_pos(3) + left_euler(3) + left_gripper(1) +
        right_pos(3) + right_euler(3) + right_gripper(1)
68         "names": {"axes": END_EFFECTOR_AXES_NAMES},
69     },
70     # Action (derived from next frame)
71     "action": {
72         "dtype": "float32",
73         "shape": (14,),
74         "names": {"axes": JOINT_AXES_NAMES},
75     },
76     # Action velocity
77     "action.velocity": {
78         "dtype": "float32",
79         "shape": (14,),
80         "names": {"axes": JOINT_AXES_NAMES},
81     },
82     # Action effort
83     "action.effort": {
84         "dtype": "float32",
85         "shape": (14,),
86         "names": {"axes": JOINT_AXES_NAMES},
87     },
88     # Action end-effector poses (flattened)
89     "action.end": {
90         "dtype": "float32",
91         "shape": (14,), # left_pos(3) + left_euler(3) + left_gripper(1) +
        right_pos(3) + right_euler(3) + right_gripper(1)
92         "names": {"axes": END_EFFECTOR_AXES_NAMES},
93     },
94     # Image features
95     "observation.images.hand_left": {
96         "dtype": "video",
97         "shape": (IMAGE_HEIGHT, IMAGE_WIDTH, IMAGE_CHANNELS),
98         "names": ["height", "width", "channels"],
99     },
100    "observation.images.hand_right": {
101        "dtype": "video",
102        "shape": (IMAGE_HEIGHT, IMAGE_WIDTH, IMAGE_CHANNELS),
103        "names": ["height", "width", "channels"],

```

```

104     },
105     "observation.images.head": {
106         "dtype": "video",
107         "shape": (IMAGE_HEIGHT, IMAGE_WIDTH, IMAGE_CHANNELS),
108         "names": ["height", "width", "channels"],
109     },
110 }
111

```

天机

代码块

```

1  TIANJI_CONFIG = {
2      "robot_type": "TIANJI",
3      "arm_dim": 14,          # l_j1~l_j7 + r_j1~r_j7
4      "effector_dim": 2,      # left_gripper + right_gripper
5      "cameras": ["head", "hand_left", "hand_right"],
6      "target_image_size": (640, 480), # (width, height)
7      "motor_names": [
8          "l_j1", "l_j2", "l_j3", "l_j4", "l_j5", "l_j6", "l_j7",
9          "left_gripper",
10         "r_j1", "r_j2", "r_j3", "r_j4", "r_j5", "r_j6", "r_j7",
11         "right_gripper",
12     ],
13 }
14
15 STATE_DIM = TIANJI_CONFIG["arm_dim"] + TIANJI_CONFIG["effector_dim"] # 16
16 ACTION_DIM = STATE_DIM

```

星尘智能s1

代码块

```

1  {
2      "codebase_version": "v2.1",
3      "robot_type": "AstriBotS1",
4      "total_episodes": 200,
5      "total_frames": 63329,
6      "total_tasks": 1,

```

```
7     "chunks_size": 1000,
8     "fps": 30,
9     "splits": {
10         "train": "0:200"
11     },
12     "data_path": "data/chunk-
{episode_chunk:03d}/episode_{episode_index:06d}.parquet",
13     "video_path": "videos/chunk-
{episode_chunk:03d}/{video_key}/episode_{episode_index:06d}.mp4",
14     "features": {
15         "observation.images.head": {
16             "dtype": "video",
17             "shape": [
18                 480,
19                 640,
20                 3
21             ],
22             "names": [
23                 "height",
24                 "width",
25                 "channels"
26             ],
27             "info": {
28                 "video.height": 480,
29                 "video.width": 640,
30                 "video.codec": "av1",
31                 "video.pix_fmt": "yuv420p",
32                 "video.is_depth_map": false,
33                 "video.fps": 30,
34                 "video.channels": 3,
35                 "has_audio": false
36             }
37         },
38         "observation.images.hand_left": {
39             "dtype": "video",
40             "shape": [
41                 480,
42                 640,
43                 3
44             ],
45             "names": [
46                 "height",
47                 "width",
48                 "channels"
49             ],
50             "info": {
51                 "video.height": 480,
```

```
52         "video.width": 640,
53         "video.codec": "av1",
54         "video.pix_fmt": "yuv420p",
55         "video.is_depth_map": false,
56         "video.fps": 30,
57         "video.channels": 3,
58         "has_audio": false
59     }
60 },
61 "observation.images.hand_right": {
62     "dtype": "video",
63     "shape": [
64         480,
65         640,
66         3
67     ],
68     "names": [
69         "height",
70         "width",
71         "channels"
72     ],
73     "info": {
74         "video.height": 480,
75         "video.width": 640,
76         "video.codec": "av1",
77         "video.pix_fmt": "yuv420p",
78         "video.is_depth_map": false,
79         "video.fps": 30,
80         "video.channels": 3,
81         "has_audio": false
82     }
83 },
84 "observation.images.torso": {
85     "dtype": "video",
86     "shape": [
87         480,
88         640,
89         3
90     ],
91     "names": [
92         "height",
93         "width",
94         "channels"
95     ],
96     "info": {
97         "video.height": 480,
98         "video.width": 640,
```



```
99         "video.codec": "av1",
100         "video.pix_fmt": "yuv420p",
101         "video.is_depth_map": false,
102         "video.fps": 30,
103         "video.channels": 3,
104         "has_audio": false
105     }
106 },
107     "observation.state": {
108         "dtype": "float32",
109         "shape": [
110             25
111         ],
112         "names": {
113             "motors": [
114                 "l_shoulder_pitch",
115                 "l_shoulder_roll",
116                 "l_shoulder_yaw",
117                 "l_elbow_pitch",
118                 "l_wrist_yaw",
119                 "l_wrist_pitch",
120                 "l_wrist_roll",
121                 "r_shoulder_pitch",
122                 "r_shoulder_roll",
123                 "r_shoulder_yaw",
124                 "r_elbow_pitch",
125                 "r_wrist_yaw",
126                 "r_wrist_pitch",
127                 "r_wrist_roll",
128                 "left_claw",
129                 "right_claw",
130                 "head_pan",
131                 "head_tilt",
132                 "ankle_pitch",
133                 "knee_pitch",
134                 "hip_pitch",
135                 "hip_yaw",
136                 "chassis_x*",
137                 "chassis_y*",
138                 "chassis_yaw*"
139             ]
140         }
141     },
142     "action": {
143         "dtype": "float32",
144         "shape": [
145             25
```

```
146         ],
147         "names": {
148             "motors": [
149                 "l_shoulder_pitch",
150                 "l_shoulder_roll",
151                 "l_shoulder_yaw",
152                 "l_elbow_pitch",
153                 "l_wrist_yaw",
154                 "l_wrist_pitch",
155                 "l_wrist_roll",
156                 "r_shoulder_pitch",
157                 "r_shoulder_roll",
158                 "r_shoulder_yaw",
159                 "r_elbow_pitch",
160                 "r_wrist_yaw",
161                 "r_wrist_pitch",
162                 "r_wrist_roll",
163                 "left_claw",
164                 "right_claw",
165                 "head_pan",
166                 "head_tilt",
167                 "ankle_pitch",
168                 "knee_pitch",
169                 "hip_pitch",
170                 "hip_yaw",
171                 "chassis_x*",
172                 "chassis_y*",
173                 "chassis_yaw*"
174             ]
175         }
176     },
177     "timestamp": {
178         "dtype": "float32",
179         "shape": [
180             1
181         ],
182         "names": null
183     },
184     "frame_index": {
185         "dtype": "int64",
186         "shape": [
187             1
188         ],
189         "names": null
190     },
191     "episode_index": {
192         "dtype": "int64",
```

```

193         "shape": [
194             1
195         ],
196         "names": null
197     },
198     "index": {
199         "dtype": "int64",
200         "shape": [
201             1
202         ],
203         "names": null
204     },
205     "task_index": {
206         "dtype": "int64",
207         "shape": [
208             1
209         ],
210         "names": null
211     }
212 },
213 "total_chunks": 1,
214 "total_videos": 800
215 }

```

fr3

代码块

```

1  {
2      "codebase_version": "v2.1",
3      "robot_type": "fr3",
4      "total_episodes": 300,
5      "total_frames": 188321,
6      "total_tasks": 1,
7      "chunks_size": 1000,
8      "fps": 30,
9      "splits": {
10         "train": "0:300"
11     },
12     "data_path": "data/chunk-
{episode_chunk:03d}/episode_{episode_index:06d}.parquet",
13     "video_path": "videos/chunk-
{episode_chunk:03d}/{video_key}/episode_{episode_index:06d}.mp4",
14     "features": {

```

```
15     "observation.images.head": {
16         "dtype": "video",
17         "shape": [
18             480,
19             640,
20             3
21         ],
22         "names": [
23             "height",
24             "width",
25             "channels"
26         ],
27         "info": {
28             "video.height": 480,
29             "video.width": 640,
30             "video.codec": "av1",
31             "video.pix_fmt": "yuv420p",
32             "video.is_depth_map": false,
33             "video.fps": 30,
34             "video.channels": 3,
35             "has_audio": false
36         }
37     },
38     "observation.images.hand_left": {
39         "dtype": "video",
40         "shape": [
41             480,
42             640,
43             3
44         ],
45         "names": [
46             "height",
47             "width",
48             "channels"
49         ],
50         "info": {
51             "video.height": 480,
52             "video.width": 640,
53             "video.codec": "av1",
54             "video.pix_fmt": "yuv420p",
55             "video.is_depth_map": false,
56             "video.fps": 30,
57             "video.channels": 3,
58             "has_audio": false
59         }
60     },
61     "observation.images.hand_right": {
```

```
62         "dtype": "video",
63         "shape": [
64             480,
65             640,
66             3
67         ],
68         "names": [
69             "height",
70             "width",
71             "channels"
72         ],
73         "info": {
74             "video.height": 480,
75             "video.width": 640,
76             "video.codec": "av1",
77             "video.pix_fmt": "yuv420p",
78             "video.is_depth_map": false,
79             "video.fps": 30,
80             "video.channels": 3,
81             "has_audio": false
82         }
83     },
84     "observation.state": {
85         "dtype": "float32",
86         "shape": [
87             30
88         ],
89         "names": {
90             "motors": [
91                 "l_fr3_joint1",
92                 "l_fr3_joint2",
93                 "l_fr3_joint3",
94                 "l_fr3_joint4",
95                 "l_fr3_joint5",
96                 "l_fr3_joint6",
97                 "l_fr3_joint7",
98                 "r_fr3_joint1",
99                 "r_fr3_joint2",
100                "r_fr3_joint3",
101                "r_fr3_joint4",
102                "r_fr3_joint5",
103                "r_fr3_joint6",
104                "r_fr3_joint7",
105                "left_gripper",
106                "right_gripper",
107                "left_x",
108                "left_y",
```

```
109         "left_z",
110         "right_x",
111         "right_y",
112         "right_z",
113         "left_qx",
114         "left_qy",
115         "left_qz",
116         "left_qw",
117         "right_qx",
118         "right_qy",
119         "right_qz",
120         "right_qw"
121     ]
122 }
123 },
124 "action": {
125     "dtype": "float32",
126     "shape": [
127         30
128     ],
129     "names": {
130         "motors": [
131             "l_fr3_joint1",
132             "l_fr3_joint2",
133             "l_fr3_joint3",
134             "l_fr3_joint4",
135             "l_fr3_joint5",
136             "l_fr3_joint6",
137             "l_fr3_joint7",
138             "r_fr3_joint1",
139             "r_fr3_joint2",
140             "r_fr3_joint3",
141             "r_fr3_joint4",
142             "r_fr3_joint5",
143             "r_fr3_joint6",
144             "r_fr3_joint7",
145             "left_gripper",
146             "right_gripper",
147             "left_x",
148             "left_y",
149             "left_z",
150             "right_x",
151             "right_y",
152             "right_z",
153             "left_qx",
154             "left_qy",
155             "left_qz",
```

```
156         "left_qw",
157         "right_qx",
158         "right_qy",
159         "right_qz",
160         "right_qw"
161     ]
162 }
163 },
164 "timestamp": {
165     "dtype": "float32",
166     "shape": [
167         1
168     ],
169     "names": null
170 },
171 "frame_index": {
172     "dtype": "int64",
173     "shape": [
174         1
175     ],
176     "names": null
177 },
178 "episode_index": {
179     "dtype": "int64",
180     "shape": [
181         1
182     ],
183     "names": null
184 },
185 "index": {
186     "dtype": "int64",
187     "shape": [
188         1
189     ],
190     "names": null
191 },
192 "task_index": {
193     "dtype": "int64",
194     "shape": [
195         1
196     ],
197     "names": null
198 }
199 },
200 "total_chunks": 1,
201 "total_videos": 900
```

Cobotmagic

代码块

```
1  {
2      "codebase_version": "v2.1",
3      "robot_type": "cobotmagic",
4      "total_episodes": 51,
5      "total_frames": 67115,
6      "total_tasks": 1,
7      "chunks_size": 1000,
8      "fps": 30,
9      "splits": {
10         "train": "0:51"
11     },
12     "data_path": "data/chunk-
{episode_chunk:03d}/episode_{episode_index:06d}.parquet",
13     "video_path": "videos/chunk-
{episode_chunk:03d}/{video_key}/episode_{episode_index:06d}.mp4",
14     "features": {
15         "observation.images.head": {
16             "dtype": "video",
17             "shape": [
18                 480,
19                 640,
20                 3
21             ],
22             "names": [
23                 "height",
24                 "width",
25                 "channels"
26             ],
27             "info": {
28                 "video.height": 480,
29                 "video.width": 640,
30                 "video.codec": "av1",
31                 "video.pix_fmt": "yuv420p",
32                 "video.is_depth_map": false,
33                 "video.fps": 30,
34                 "video.channels": 3,
```



```
35         "has_audio": false
36     }
37 },
38 "observation.images.hand_left": {
39     "dtype": "video",
40     "shape": [
41         480,
42         640,
43         3
44     ],
45     "names": [
46         "height",
47         "width",
48         "channels"
49     ],
50     "info": {
51         "video.height": 480,
52         "video.width": 640,
53         "video.codec": "av1",
54         "video.pix_fmt": "yuv420p",
55         "video.is_depth_map": false,
56         "video.fps": 30,
57         "video.channels": 3,
58         "has_audio": false
59     }
60 },
61 "observation.images.hand_right": {
62     "dtype": "video",
63     "shape": [
64         480,
65         640,
66         3
67     ],
68     "names": [
69         "height",
70         "width",
71         "channels"
72     ],
73     "info": {
74         "video.height": 480,
75         "video.width": 640,
76         "video.codec": "av1",
77         "video.pix_fmt": "yuv420p",
78         "video.is_depth_map": false,
79         "video.fps": 30,
80         "video.channels": 3,
81         "has_audio": false
```

```
82         }
83     },
84     "observation.state": {
85         "dtype": "float32",
86         "shape": [
87             20
88         ],
89         "names": {
90             "motors": [
91                 "l-j1",
92                 "l-j2",
93                 "l-j3",
94                 "l-j4",
95                 "l-j5",
96                 "l-j6",
97                 "r-j1",
98                 "r-j2",
99                 "r-j3",
100                "r-j4",
101                "r-j5",
102                "r-j6",
103                "left_gripper",
104                "right_gripper",
105                "robot_angular_x",
106                "robot_angular_y",
107                "robot_angular_z",
108                "robot_vel_x",
109                "robot_vel_y",
110                "robot_vel_z"
111            ]
112        }
113     },
114     "action": {
115         "dtype": "float32",
116         "shape": [
117             20
118         ],
119         "names": {
120             "motors": [
121                 "l-j1",
122                 "l-j2",
123                 "l-j3",
124                 "l-j4",
125                 "l-j5",
126                 "l-j6",
127                 "r-j1",
128                 "r-j2",
```

```
129         "r-j3",
130         "r-j4",
131         "r-j5",
132         "r-j6",
133         "left_gripper",
134         "right_gripper",
135         "robot_angular_x",
136         "robot_angular_y",
137         "robot_angular_z",
138         "robot_vel_x",
139         "robot_vel_y",
140         "robot_vel_z"
141     ]
142 }
143 },
144 "timestamp": {
145     "dtype": "float32",
146     "shape": [
147         1
148     ],
149     "names": null
150 },
151 "frame_index": {
152     "dtype": "int64",
153     "shape": [
154         1
155     ],
156     "names": null
157 },
158 "episode_index": {
159     "dtype": "int64",
160     "shape": [
161         1
162     ],
163     "names": null
164 },
165 "index": {
166     "dtype": "int64",
167     "shape": [
168         1
169     ],
170     "names": null
171 },
172 "task_index": {
173     "dtype": "int64",
174     "shape": [
175         1
```

```

176         ],
177         "names": null
178     }
179 },
180 "total_chunks": 1,
181 "total_videos": 153
182 }

```

gr2

代码块

```

1  {
2      "codebase_version": "v2.1",
3      "robot_type": "GR2",
4      "total_episodes": 301,
5      "total_frames": 131197,
6      "total_tasks": 1,
7      "chunks_size": 1000,
8      "fps": 30,
9      "splits": {
10         "train": "0:301"
11     },
12     "data_path": "data/chunk-
{episode_chunk:03d}/episode_{episode_index:06d}.parquet",
13     "video_path": "videos/chunk-
{episode_chunk:03d}/{video_key}/episode_{episode_index:06d}.mp4",
14     "features": {
15         "observation.images.head_left": {
16             "dtype": "video",
17             "shape": [
18                 480,
19                 640,
20                 3
21             ],
22             "names": [
23                 "height",
24                 "width",
25                 "channels"
26             ],
27             "info": {
28                 "video.height": 480,
29                 "video.width": 640,
30                 "video.codec": "av1",
31                 "video.pix_fmt": "yuv420p",
32                 "video.is_depth_map": false,

```

```
33         "video.fps": 30,
34         "video.channels": 3,
35         "has_audio": false
36     }
37 },
38 "observation.images.head_right": {
39     "dtype": "video",
40     "shape": [
41         480,
42         640,
43         3
44     ],
45     "names": [
46         "height",
47         "width",
48         "channels"
49     ],
50     "info": {
51         "video.height": 480,
52         "video.width": 640,
53         "video.codec": "av1",
54         "video.pix_fmt": "yuv420p",
55         "video.is_depth_map": false,
56         "video.fps": 30,
57         "video.channels": 3,
58         "has_audio": false
59     }
60 },
61 "observation.state": {
62     "dtype": "float32",
63     "shape": [
64         41
65     ],
66     "names": {
67         "motors": [
68             "left_shoulder_pitch_joint",
69             "left_shoulder_roll_joint",
70             "left_shoulder_yaw_joint",
71             "left_elbow_pitch_joint",
72             "left_wrist_yaw_joint",
73             "left_wrist_pitch_joint",
74             "left_wrist_roll_joint",
75             "right_shoulder_pitch_joint",
76             "right_shoulder_roll_joint",
77             "right_shoulder_yaw_joint",
78             "right_elbow_pitch_joint",
79             "right_wrist_yaw_joint",
```

```

80         "right_wrist_pitch_joint",
81         "right_wrist_roll_joint",
82         "L_pinky_proximal_joint",
83         "L_ring_proximal_joint",
84         "L_middle_proximal_joint",
85         "L_index_proximal_joint",
86         "L_thumb_proximal_pitch_joint",
87         "L_thumb_proximal_yaw_joint",
88         "R_pinky_proximal_joint",
89         "R_ring_proximal_joint",
90         "R_middle_proximal_joint",
91         "R_index_proximal_joint",
92         "R_thumb_proximal_pitch_joint",
93         "R_thumb_proximal_yaw_joint",
94         "head_yaw_joint",
95         "head_pitch_joint",
96         "left_hip_pitch_joint",
97         "left_hip_roll_joint",
98         "left_hip_yaw_joint",
99         "left_knee_pitch_joint",
100        "left_ankle_roll_joint",
101        "left_ankle_pitch_joint",
102        "right_hip_pitch_joint",
103        "right_hip_roll_joint",
104        "right_hip_yaw_joint",
105        "right_knee_pitch_joint",
106        "right_ankle_roll_joint",
107        "right_ankle_pitch_joint",
108        "waist_yaw_joint"
109    ]
110 }
111 },
112 "action": {
113     "dtype": "float32",
114     "shape": [
115         41
116     ],
117     "names": {
118         "motors": [
119             "left_shoulder_pitch_joint",
120             "left_shoulder_roll_joint",
121             "left_shoulder_yaw_joint",
122             "left_elbow_pitch_joint",
123             "left_wrist_yaw_joint",
124             "left_wrist_pitch_joint",
125             "left_wrist_roll_joint",
126             "right_shoulder_pitch_joint",

```

```
127         "right_shoulder_roll_joint",
128         "right_shoulder_yaw_joint",
129         "right_elbow_pitch_joint",
130         "right_wrist_yaw_joint",
131         "right_wrist_pitch_joint",
132         "right_wrist_roll_joint",
133         "L_pinky_proximal_joint",
134         "L_ring_proximal_joint",
135         "L_middle_proximal_joint",
136         "L_index_proximal_joint",
137         "L_thumb_proximal_pitch_joint",
138         "L_thumb_proximal_yaw_joint",
139         "R_pinky_proximal_joint",
140         "R_ring_proximal_joint",
141         "R_middle_proximal_joint",
142         "R_index_proximal_joint",
143         "R_thumb_proximal_pitch_joint",
144         "R_thumb_proximal_yaw_joint",
145         "head_yaw_joint",
146         "head_pitch_joint",
147         "left_hip_pitch_joint",
148         "left_hip_roll_joint",
149         "left_hip_yaw_joint",
150         "left_knee_pitch_joint",
151         "left_ankle_roll_joint",
152         "left_ankle_pitch_joint",
153         "right_hip_pitch_joint",
154         "right_hip_roll_joint",
155         "right_hip_yaw_joint",
156         "right_knee_pitch_joint",
157         "right_ankle_roll_joint",
158         "right_ankle_pitch_joint",
159         "waist_yaw_joint"
160     ]
161 }
162 },
163 "timestamp": {
164     "dtype": "float32",
165     "shape": [
166         1
167     ],
168     "names": null
169 },
170 "frame_index": {
171     "dtype": "int64",
172     "shape": [
173         1
```

```

174         ],
175         "names": null
176     },
177     "episode_index": {
178         "dtype": "int64",
179         "shape": [
180             1
181         ],
182         "names": null
183     },
184     "index": {
185         "dtype": "int64",
186         "shape": [
187             1
188         ],
189         "names": null
190     },
191     "task_index": {
192         "dtype": "int64",
193         "shape": [
194             1
195         ],
196         "names": null
197     }
198 },
199 "total_chunks": 1,
200 "total_videos": 602
201 }

```

星海图R1

代码块

```

1  # 星海图 R1 机器人配置
2  R1_CONFIG = {
3      "robot_type": "xinghaitu_r1",
4      "arm_dim": 12,          # 左臂6 + 右臂6
5      "gripper_dim": 2,      # 左夹爪 + 右夹爪
6      "cameras": ["head", "hand_left", "hand_right"],
7      "image_shape": (480, 640, 3),
8  }
9
10 # State 和 Action 维度: 12 (arm) + 2 (gripper) = 14
11 STATE_DIM = R1_CONFIG["arm_dim"] + R1_CONFIG["gripper_dim"]

```


乐聚

代码块

```

1  LEJU_CONFIG = {
2      "robot_type": "lejukuafu",
3      "arm_dim": 14,          # 左臂7 + 右臂7
4      "effector_dim": 2,      # 左夹爪 + 右夹爪
5      "head_dim": 2,          # head_yaw + head_pitch
6      "leg_dim": 12,          # 左腿6 + 右腿6
7      "cameras": ["head", "hand_left", "hand_right"],
8      "target_image_size": (640, 480), # (width, height) - 统一resize目标尺寸: 宽
        640、高480
9  }
10
11  # State/Action: arm(14) + effector(2) + head(2) + leg(12) = 30
12  STATE_DIM = LEJU_CONFIG["arm_dim"] + LEJU_CONFIG["effector_dim"] +
    LEJU_CONFIG["head_dim"] + LEJU_CONFIG["leg_dim"]
13  ACTION_DIM = STATE_DIM
14
15  # 电机名称定义 (与参考JSON一致)
16  STATE_NAMES = [
17      "l_arm_pitch", "l_arm_roll", "l_arm_yaw", "l_forearm_pitch",
18      "l_hand_yaw", "l_hand_pitch", "l_hand_roll",
19      "r_arm_pitch", "r_arm_roll", "r_arm_yaw", "r_forearm_pitch",
20      "r_hand_yaw", "r_hand_pitch", "r_hand_roll",
21      "left_claw", "right_claw",
22      "head_yaw", "head_pitch",
23      "l_leg_roll", "l_leg_yaw", "l_leg_pitch", "l_knee", "l_foot_pitch",
        "l_foot_roll",
24      "r_leg_roll", "r_leg_yaw", "r_leg_pitch", "r_knee", "r_foot_pitch",
        "r_foot_roll"
25  ]
26
27  ACTION_NAMES = STATE_NAMES # 动作名称与状态名称相同
28

```

青龙ROS1

代码块

```

1  QINGLONG_CONFIG = {

```

```

2     "robot_type": "QinLongROS1",
3     "arm_dim": 14,           # 左臂7 + 右臂7
4     "effector_dim": 2,      # 左夹爪 + 右夹爪
5     "cameras": ["head", "hand_left", "hand_right"],
6     "target_image_size": (640, 480), # (width, height) - 统一resize目标尺寸: 宽
640、高480
7     "motor_names": [
8         "J_arm_l_01", "J_arm_l_02", "J_arm_l_03", "J_arm_l_04", "J_arm_l_05",
9         "J_arm_l_06", "J_arm_l_07",
10        "J_arm_r_01", "J_arm_r_02", "J_arm_r_03", "J_arm_r_04", "J_arm_r_05",
11        "J_arm_r_06", "J_arm_r_07",
12        "left_gripper", "right_gripper"
13    ]
14 }
15
16 STATE_DIM = QINGLONG_CONFIG["arm_dim"] + QINGLONG_CONFIG["effector_dim"] # 16
17 ACTION_DIM = STATE_DIM
18

```

青龙ROS2

代码块

```

1  QINGLONG_CONFIG = {
2      "robot_type": "QinLongROS2",
3      "arm_dim": 14,
4      "effector_dim": 2,
5      "head_dim": 2,
6      "waist_dim": 3,
7      "leg_dim": 12,
8      "cameras": ["head", "hand_left", "hand_right"],
9      "target_image_size": (640, 480), # (width, height)
10     "motor_names": [
11         # arm (14)
12         "J_arm_l_01", "J_arm_l_02", "J_arm_l_03", "J_arm_l_04", "J_arm_l_05",
13         "J_arm_l_06", "J_arm_l_07",
14         "J_arm_r_01", "J_arm_r_02", "J_arm_r_03", "J_arm_r_04", "J_arm_r_05",
15         "J_arm_r_06", "J_arm_r_07",
16         # effector (2)
17         "left_gripper", "right_gripper",
18         # head (2)
19         "J_head_yaw", "J_head_pitch",
20         # waist (3)
21         "J_waist_pitch", "J_waist_roll", "J_waist_yaw",
22         # leg (12)

```

```

21     "J_hip_l_roll", "J_hip_l_yaw", "J_hip_l_pitch", "J_knee_l_pitch",
    "J_ankle_l_pitch", "J_ankle_l_roll",
22     "J_hip_r_roll", "J_hip_r_yaw", "J_hip_r_pitch", "J_knee_r_pitch",
    "J_ankle_r_pitch", "J_ankle_r_roll",
23 ],
24 }
25
26 STATE_DIM = (
27     QINGLONG_CONFIG["arm_dim"]
28     + QINGLONG_CONFIG["effector_dim"]
29     + QINGLONG_CONFIG["head_dim"]
30     + QINGLONG_CONFIG["waist_dim"]
31     + QINGLONG_CONFIG["leg_dim"]
32 ) # 33
33 ACTION_DIM = STATE_DIM

```

UR5e

代码块

```

1  UR5E_CONFIG = {
2      "robot_type": "DualUR5e",
3      "arm_dim": 12,          # 6 + 6
4      "effector_dim": 2,     # 左右夹爪
5      "cameras": ["head", "hand_left", "hand_right"],
6      "target_image_size": (640, 480), # (width, height)
7      "effector_clip": (0.0, 100.0),   # 夹爪开合度范围
8      "motor_names": [
9          # left UR5e (6)
10         "l_shoulder_pan_joint", "l_shoulder_lift_joint", "l_elbow_joint",
11         "l_wrist_1_joint", "l_wrist_2_joint", "l_wrist_3_joint",
12         # right UR5e (6)
13         "r_shoulder_pan_joint", "r_shoulder_lift_joint", "r_elbow_joint",
14         "r_wrist_1_joint", "r_wrist_2_joint", "r_wrist_3_joint",
15         # gripper (2)
16         "left_gripper", "right_gripper",
17     ],
18 }
19
20 STATE_DIM = UR5E_CONFIG["arm_dim"] + UR5E_CONFIG["effector_dim"] # 14
21 ACTION_DIM = STATE_DIM

```

Dwheel

代码块

```
DWHEEL_CONFIG = {
2     "robot_type": "Dwheel",
3     "arm_dim": 14,
4     "effector_dim": 2,
5     "waist_dim": 2,
6     "robot_dim": 2,
7     "cameras": ["head", "hand_left", "hand_right"],
8     "target_image_size": (640, 480), # (width, height)
9     "motor_names": [
10         "l-j1",
11         "l-j2",
12         "l-j3",
13         "l-j4",
14         "l-j5",
15         "l-j6",
16         "l-j7",
17         "r-j1",
18         "r-j2",
19         "r-j3",
20         "r-j4",
21         "r-j5",
22         "r-j6",
23         "r-j7",
24         "left_gripper",
25         "right_gripper",
26         "J_waist_pitch",
27         "J_waist_lift",
28         "base_vx",
29         "base_vyaw",
30     ],
31 }
32
33 STATE_DIM = (
34     DWHEEL_CONFIG["arm_dim"]
35     + DWHEEL_CONFIG["effector_dim"]
36     + DWHEEL_CONFIG["waist_dim"]
37     + DWHEEL_CONFIG["robot_dim"]
38 ) # 20
39 ACTION_DIM = STATE_DIM
```

Robomind

agilex_3cam

代码块

```

1  CONFIG = {
2      "robot_type": "agilex_3cam",
3      "arm_dim": 12,          # 左臂6 + 右臂6
4      "effector_dim": 2,      # 左爪 + 右爪 (prismatic 0~0.07 m)
5      "cameras": ["camera_front", "camera_left", "camera_right"],
6      "target_image_size": (640, 480),
7      "motor_names": [
8          "arm_master_l_joint1",
9          "arm_master_l_joint2",
10         "arm_master_l_joint3",
11         "arm_master_l_joint4",
12         "arm_master_l_joint5",
13         "arm_master_l_joint6",
14         "left_gripper",
15         "arm_master_r_joint1",
16         "arm_master_r_joint2",
17         "arm_master_r_joint3",
18         "arm_master_r_joint4",
19         "arm_master_r_joint5",
20         "arm_master_r_joint6",
21         "right_gripper",
22     ],
23 }
24
25 STATE_DIM = CONFIG["arm_dim"] + CONFIG["effector_dim"] # 14
26 ACTION_DIM = STATE_DIM

```

agilex_1cam

代码块

```

1  CONFIG = {
2      "robot_type": "agilex_1cam",
3      "arm_dim": 12,
4      "effector_dim": 2,
5      "cameras": ["camera_front"],
6      "target_image_size": (640, 480),
7      "motor_names": [
8          "arm_master_l_joint1", "arm_master_l_joint2", "arm_master_l_joint3",
9          "arm_master_l_joint4", "arm_master_l_joint5", "arm_master_l_joint6",
10         "left_gripper",
11         "arm_master_r_joint1", "arm_master_r_joint2", "arm_master_r_joint3",
12         "arm_master_r_joint4", "arm_master_r_joint5", "arm_master_r_joint6",
13         "right_gripper",
14     ],
15 }

```

```

16
17 STATE_DIM = 14
18 ACTION_DIM = STATE_DIM
19
20 TASKS = [
21     "flip_cup_and_place_on_plate_with_arms",           # 461 ep
22     "place_corn_on_plate_with_both_arms_1cam",         # 54 ep
23     "stack_green_and_blue_bowls_with_arms_1cam",       # 183 ep
24 ]

```

agilex_mobile

代码块

```

1  CONFIG = {
2      "robot_type": "agilex_mobile",
3      "arm_dim": 12,
4      "effector_dim": 2,
5      "cameras": ["camera_front", "camera_left", "camera_right"],
6      "target_image_size": (640, 480),
7      "motor_names": [
8          "arm_master_l_joint1", "arm_master_l_joint2", "arm_master_l_joint3",
9          "arm_master_l_joint4", "arm_master_l_joint5", "arm_master_l_joint6",
10         "left_gripper",
11         "arm_master_r_joint1", "arm_master_r_joint2", "arm_master_r_joint3",
12         "arm_master_r_joint4", "arm_master_r_joint5", "arm_master_r_joint6",
13         "right_gripper",
14     ],
15 }
16
17 STATE_DIM = 14
18 ACTION_DIM = STATE_DIM

```

franka_6cam

代码块

```

1  CONFIG = {
2      "robot_type": "franka_6cam",
3      "arm_dim": 16,           # 左臂8 + 右臂8 (arm_includes_gripper=True, 第8维
                                # =gripper)
4      "effector_dim": 0,      # 不单独叠加
5      "cameras": [
6          "camera_front", "camera_left", "camera_right",

```

```

7         "camera_top", "camera_wrist_left", "camera_wrist_right",
8     ],
9     "target_image_size": (640, 480),
10    "motor_names": [
11        "l_franka_joint1", "l_franka_joint2", "l_franka_joint3",
12        "l_franka_joint4", "l_franka_joint5", "l_franka_joint6",
13        "l_franka_joint7", "l_franka_gripper",
14        "r_franka_joint1", "r_franka_joint2", "r_franka_joint3",
15        "r_franka_joint4", "r_franka_joint5", "r_franka_joint6",
16        "r_franka_joint7", "r_franka_gripper",
17    ],
18 }
19
20 STATE_DIM = CONFIG["arm_dim"] # 16
21 ACTION_DIM = STATE_DIM

```

franka_4cam

代码块

```

1  CONFIG = {
2      "robot_type": "franka_4cam",
3      "arm_dim": 16,
4      "effector_dim": 0,
5      "cameras": ["camera_front", "camera_left", "camera_right", "camera_top"],
6      "target_image_size": (640, 480),
7      "motor_names": [
8          "l_franka_joint1", "l_franka_joint2", "l_franka_joint3",
9          "l_franka_joint4", "l_franka_joint5", "l_franka_joint6",
10         "l_franka_joint7", "l_franka_gripper",
11         "r_franka_joint1", "r_franka_joint2", "r_franka_joint3",
12         "r_franka_joint4", "r_franka_joint5", "r_franka_joint6",
13         "r_franka_joint7", "r_franka_gripper",
14     ],
15 }
16
17 STATE_DIM = 16
18 ACTION_DIM = STATE_DIM

```

franka_3cam

代码块

```

1  CONFIG = {

```

```

2     "robot_type": "franka_3cam",
3     "arm_dim": 16,
4     "effector_dim": 0,
5     "cameras": ["camera_front", "camera_left", "camera_right"],
6     "target_image_size": (640, 480),
7     "motor_names": [
8         "l_franka_joint1", "l_franka_joint2", "l_franka_joint3",
9         "l_franka_joint4", "l_franka_joint5", "l_franka_joint6",
10        "l_franka_joint7", "l_franka_gripper",
11        "r_franka_joint1", "r_franka_joint2", "r_franka_joint3",
12        "r_franka_joint4", "r_franka_joint5", "r_franka_joint6",
13        "r_franka_joint7", "r_franka_gripper",
14    ],
15 }
16
17 TASKS = ["open_drawer_take_button_close_drawer_with_arms_3cam"] # 1 ep

```

franka_1cam

代码块

```

1  CONFIG = {
2      "robot_type": "franka_1cam",
3      "arm_dim": 16,
4      "effector_dim": 0,
5      "cameras": ["camera_front"],
6      "target_image_size": (640, 480),
7      "motor_names": [
8          "l_franka_joint1", "l_franka_joint2", "l_franka_joint3",
9          "l_franka_joint4", "l_franka_joint5", "l_franka_joint6",
10         "l_franka_joint7", "l_franka_gripper",
11         "r_franka_joint1", "r_franka_joint2", "r_franka_joint3",
12         "r_franka_joint4", "r_franka_joint5", "r_franka_joint6",
13         "r_franka_joint7", "r_franka_gripper",
14     ],
15 }
16
17 TASKS = ["open_drawer_take_button_close_drawer_with_arms_1cam"] # 1 ep

```

tianyi

代码块

```

1  CONFIG = {

```



```

2     "robot_type": "tianyiy",
3     "arm_dim": 14,          # 左臂7 + 右臂7
4     "effector_dim": 2,      # 左爪 + 右爪 (归一化 [0, 1])
5     "cameras": ["camera_top"],
6     "target_image_size": (640, 480),
7     "motor_names": [
8         "l_arm_joint1", "l_arm_joint2", "l_arm_joint3", "l_arm_joint4",
9         "l_arm_joint5", "l_arm_joint6", "l_arm_joint7",
10        "left_gripper",
11        "r_arm_joint1", "r_arm_joint2", "r_arm_joint3", "r_arm_joint4",
12        "r_arm_joint5", "r_arm_joint6", "r_arm_joint7",
13        "right_gripper",
14    ],
15 }
16
17 STATE_DIM = 16
18 ACTION_DIM = STATE_DIM

```

tianyiy_mobile

代码块

```

1  CONFIG = {
2      "robot_type": "tianyiy_mobile",
3      "arm_dim": 14,
4      "effector_dim": 2,
5      "cameras": ["camera_top"],
6      "target_image_size": (640, 480),
7      "motor_names": [
8          "l_arm_joint1", "l_arm_joint2", "l_arm_joint3", "l_arm_joint4",
9          "l_arm_joint5", "l_arm_joint6", "l_arm_joint7",
10         "left_gripper",
11         "r_arm_joint1", "r_arm_joint2", "r_arm_joint3", "r_arm_joint4",
12         "r_arm_joint5", "r_arm_joint6", "r_arm_joint7",
13         "right_gripper",
14     ],
15 }
16
17 STATE_DIM = 16
18 ACTION_DIM = STATE_DIM

```

tienkung_dim16

```

1  CONFIG = {
2      "robot_type": "tienkung_dim16",
3      "schema": "dim_16",
4      "arm_dim": 14,          # 左臂7 + 右臂7
5      "effector_dim": 2,      # 左爪 + 右爪
6      "cameras": ["camera_top"],
7      "target_image_size": (640, 480),
8      "motor_names": [
9          "l_arm_joint1", "l_arm_joint2", "l_arm_joint3", "l_arm_joint4",
10         "l_arm_joint5", "l_arm_joint6", "l_arm_joint7",
11         "left_gripper",
12         "r_arm_joint1", "r_arm_joint2", "r_arm_joint3", "r_arm_joint4",
13         "r_arm_joint5", "r_arm_joint6", "r_arm_joint7",
14         "right_gripper",
15     ],
16 }
17
18 STATE_DIM = 16
19 ACTION_DIM = STATE_DIM

```

tienkung_dim26

代码块

```

1  CONFIG = {
2      "robot_type": "tienkung_dim26",
3      "schema": "dim_26",
4      "arm_dim": 14,          # 左臂7 + 右臂7
5      "finger_dim": 12,       # 左手6 + 右手6 (灵巧手, mm 0~100)
6      "cameras": ["camera_top"],
7      "target_image_size": (640, 480),
8      "motor_names": [
9          "l_arm_joint1", "l_arm_joint2", "l_arm_joint3", "l_arm_joint4",
10         "l_arm_joint5", "l_arm_joint6", "l_arm_joint7",
11         "left_thumb_0", "left_thumb_1", "left_index",
12         "left_middle", "left_ring", "left_pinky",
13         "r_arm_joint1", "r_arm_joint2", "r_arm_joint3", "r_arm_joint4",
14         "r_arm_joint5", "r_arm_joint6", "r_arm_joint7",
15         "right_thumb_0", "right_thumb_1", "right_index",
16         "right_middle", "right_ring", "right_pinky",
17     ],
18 }
19
20 STATE_DIM = CONFIG["arm_dim"] + CONFIG["finger_dim"] # 26
21 ACTION_DIM = STATE_DIM
22

```

```

23 TASKS = [
24     "flip_black_knob_to_close_circuit",      # 300 ep
25     "open_door_with_both_hands",            # 300 ep
26     "press_green_start_button",              # 300 ep
27     "release_door_handle",                   # 300 ep
28     "rotate_and_turn_off_gray_switch",        # 195 ep
29     "trip_black_knob_to_open_circuit",        # 300 ep
30     "turn_on_light_switch_1",                 # 300 ep
31 ]

```

ur5

代码块

```

1  CONFIG = {
2      "robot_type": "ur5",
3      "arm_dim": 12,          # 左 UR5 (6) + 右 UR5 (6)
4      "effector_dim": 2,      # 左夹爪 + 右夹爪 (归一化 [0, 1])
5      "cameras": [
6          "camera_front", "camera_left", "camera_right",
7          "camera_top", "camera_wrist_left", "camera_wrist_right",
8      ],
9      "target_image_size": (640, 480),
10     "motor_names": [
11         "l_shoulder_pan_joint", "l_shoulder_lift_joint", "l_elbow_joint",
12         "l_wrist_1_joint", "l_wrist_2_joint", "l_wrist_3_joint",
13         "left_gripper",
14         "r_shoulder_pan_joint", "r_shoulder_lift_joint", "r_elbow_joint",
15         "r_wrist_1_joint", "r_wrist_2_joint", "r_wrist_3_joint",
16         "right_gripper",
17     ],
18 }
19
20 STATE_DIM = CONFIG["arm_dim"] + CONFIG["effector_dim"] # 14
21 ACTION_DIM = STATE_DIM

```

ur5_dex灵巧手

代码块

```

1  CONFIG = {
2      "robot_type": "ur5_dex",
3      "arm_dim": 12,          # 左 UR5 (6) + 右 UR5 (6)
4      "finger_dim": 24,       # 左手 12 + 右手 12 (灵巧手, mm 0~100)

```

```

5     "cameras": [
6         "camera_front", "camera_left", "camera_right",
7         "camera_top", "camera_wrist_left", "camera_wrist_right",
8     ],
9     "target_image_size": (640, 480),
10    "motor_names": [
11        "l_shoulder_pan_joint", "l_shoulder_lift_joint", "l_elbow_joint",
12        "l_wrist_1_joint", "l_wrist_2_joint", "l_wrist_3_joint",
13        "l_finger_0", "l_finger_1", "l_finger_2", "l_finger_3",
14        "l_finger_4", "l_finger_5", "l_finger_6", "l_finger_7",
15        "l_finger_8", "l_finger_9", "l_finger_10", "l_finger_11",
16        "r_shoulder_pan_joint", "r_shoulder_lift_joint", "r_elbow_joint",
17        "r_wrist_1_joint", "r_wrist_2_joint", "r_wrist_3_joint",
18        "r_finger_0", "r_finger_1", "r_finger_2", "r_finger_3",
19        "r_finger_4", "r_finger_5", "r_finger_6", "r_finger_7",
20        "r_finger_8", "r_finger_9", "r_finger_10", "r_finger_11",
21    ],
22 }
23
24 STATE_DIM = CONFIG["arm_dim"] + CONFIG["finger_dim"] # 36
25 ACTION_DIM = STATE_DIM
26
27 TASKS = [
28     "box_in_the_basket",
29     "food_into_the_basket",
30     "general_objects_grasping_task",
31     "pick_cup_into_the_box_right (speed 100)_20250520",
32     "pour_water_task",
33     "tennis_throwing",
34 ]

```

sim_libero_franka

代码块

```

1  {
2      "state": {
3          "joint": {
4              "start": 0,
5              "end": 7
6          },
7      "eef": {

```

```
8         "start": 7,
9         "end": 13
10    },
11    "gripper": {
12        "start": 13,
13        "end": 15
14    },
15    "reward": {
16        "start": 15,
17        "end": 16
18    }
19 },
20 "action": {
21     "eef": {
22         "start": 0,
23         "end": 6
24     },
25     "gripper": {
26         "start": 6,
27         "end": 7
28     }
29 },
30 "video": {
31     "image": {
32         "original_key": "observation.images.front"
33     },
34     "wrist_image": {
35         "original_key": "observation.images.wrist"
36     }
37 },
38 "annotation": {
39     "human.task_description": {
40         "original_key": "task_index"
41     }
42 }
43 }
```

sim_robotwin2_franka

```
1  {
2    "state": {
3      "left_arm_qpos": {
4        "start": 0,
5        "end": 7
6      },
7      "left_arm_gripper": {
8        "start": 7,
9        "end": 8
10     },
11     "right_arm_qpos": {
12       "start": 8,
13       "end": 15
14     },
15     "right_arm_gripper": {
16       "start": 15,
17       "end": 16
18     },
19     "left_arm_eef": {
20       "start": 16,
21       "end": 23
22     },
23     "right_arm_eef": {
24       "start": 23,
25       "end": 30
26   }
27 },
28 "action": {
29   "left_arm_qpos": {
30     "start": 0,
31     "end": 7
32   },
33   "left_arm_gripper": {
34     "start": 7,
35     "end": 8
36   },
37   "right_arm_qpos": {
38     "start": 8,
39     "end": 15
40   },
41   "right_arm_gripper": {
42     "start": 15,
43     "end": 16
44   },
45   "left_arm_eef": {
46     "start": 16,
47     "end": 23
```

```

48     },
49     "right_arm_eef": {
50         "start": 23,
51         "end": 30
52     }
53 },
54 "video": {
55     "cam_high": {
56         "original_key": "observation.images.head_camera"
57     },
58     "cam_left_wrist": {
59         "original_key": "observation.images.left_camera"
60     },
61     "cam_right_wrist": {
62         "original_key": "observation.images.right_camera"
63     },
64     "cam_third_wrist": {
65         "original_key": "observation.images.third_view"
66     }
67 },
68 "annotation": {
69     "human.task_description": {
70         "original_key": "task_index"
71     }
72 }
73 }
74

```

sim_robotwin2_ur5

代码块

```

1
2 {
3     "state": {
4         "left_arm_qpos": {
5             "start": 0,
6             "end": 6
7         },
8         "left_arm_gripper": {
9             "start": 6,
10            "end": 7
11        },
12        "right_arm_qpos": {

```

```
13     "start": 7,
14     "end": 13
15 },
16 "right_arm_gripper": {
17     "start": 13,
18     "end": 14
19 },
20 "left_arm_eef": {
21     "start": 14,
22     "end": 21
23 },
24 "right_arm_eef": {
25     "start": 21,
26     "end": 28
27 }
28 },
29 "action": {
30     "left_arm_qpos": {
31         "start": 0,
32         "end": 6
33     },
34     "left_arm_gripper": {
35         "start": 6,
36         "end": 7
37     },
38     "right_arm_qpos": {
39         "start": 7,
40         "end": 13
41     },
42     "right_arm_gripper": {
43         "start": 13,
44         "end": 14
45     },
46     "left_arm_eef": {
47         "start": 14,
48         "end": 21
49     },
50     "right_arm_eef": {
51         "start": 21,
52         "end": 28
53     }
54 },
55 "video": {
56     "cam_high": {
57         "original_key": "observation.images.head_camera"
58     },
59     "cam_left_wrist": {
```



```
60     "original_key": "observation.images.left_camera"
61 },
62     "cam_right_wrist": {
63         "original_key": "observation.images.right_camera"
64     },
65     "cam_third_wrist": {
66         "original_key": "observation.images.third_view"
67     }
68 },
69     "annotation": {
70         "human.task_description": {
71             "original_key": "task_index"
72         }
73     }
74 }
```

sim_robotwin2_aloha_agilex

代码块

```
1
2 {
3     "state": {
4         "left_arm_qpos": {
5             "start": 0,
6             "end": 6
7         },
8         "left_arm_gripper": {
9             "start": 6,
10            "end": 7
11        },
12        "right_arm_qpos": {
13            "start": 7,
14            "end": 13
15        },
16        "right_arm_gripper": {
17            "start": 13,
18            "end": 14
19        },
20        "left_arm_eef": {
21            "start": 14,
22            "end": 21
```

```
23     },
24     "right_arm_eef": {
25         "start": 21,
26         "end": 28
27     }
28 },
29 "action": {
30     "left_arm_qpos": {
31         "start": 0,
32         "end": 6
33     },
34     "left_arm_gripper": {
35         "start": 6,
36         "end": 7
37     },
38     "right_arm_qpos": {
39         "start": 7,
40         "end": 13
41     },
42     "right_arm_gripper": {
43         "start": 13,
44         "end": 14
45     },
46     "left_arm_eef": {
47         "start": 14,
48         "end": 21
49     },
50     "right_arm_eef": {
51         "start": 21,
52         "end": 28
53     }
54 },
55 "video": {
56     "cam_high": {
57         "original_key": "observation.images.head_camera"
58     },
59     "cam_left_wrist": {
60         "original_key": "observation.images.left_camera"
61     },
62     "cam_right_wrist": {
63         "original_key": "observation.images.right_camera"
64     },
65     "cam_third_wrist": {
66         "original_key": "observation.images.front_camera"
67     }
68 },
69 "annotation": {
```

```
70     "human.task_description": {
71         "original_key": "task_index"
72     }
73 }
74 }
```

sim_robotwin2_arx_x5

代码块

```
1
2 {
3     "state": {
4         "left_arm_qpos": {
5             "start": 0,
6             "end": 6
7         },
8         "left_arm_gripper": {
9             "start": 6,
10            "end": 7
11        },
12        "right_arm_qpos": {
13            "start": 7,
14            "end": 13
15        },
16        "right_arm_gripper": {
17            "start": 13,
18            "end": 14
19        },
20        "left_arm_eef": {
21            "start": 14,
22            "end": 21
23        },
24        "right_arm_eef": {
25            "start": 21,
26            "end": 28
27        }
28    },
29    "action": {
30        "left_arm_qpos": {
31            "start": 0,
32            "end": 6
33        },
34        "left_arm_gripper": {
35            "start": 6,
36            "end": 7
```

```

37     },
38     "right_arm_qpos": {
39         "start": 7,
40         "end": 13
41     },
42     "right_arm_gripper": {
43         "start": 13,
44         "end": 14
45     },
46     "left_arm_eef": {
47         "start": 14,
48         "end": 21
49     },
50     "right_arm_eef": {
51         "start": 21,
52         "end": 28
53     }
54 },
55 "video": {
56     "cam_high": {
57         "original_key": "observation.images.head_camera"
58     },
59     "cam_left_wrist": {
60         "original_key": "observation.images.left_camera"
61     },
62     "cam_right_wrist": {
63         "original_key": "observation.images.right_camera"
64     },
65 },
66 "annotation": {
67     "human.task_description": {
68         "original_key": "task_index"
69     }
70 }
71 }

```

sim_robotwin2_piper

代码块

```

1
2 {
3     "state": {
4         "left_arm_qpos": {

```

```
5     "start": 0,
6     "end": 6
7 },
8 "left_arm_gripper": {
9     "start": 6,
10    "end": 7
11 },
12 "right_arm_qpos": {
13     "start": 7,
14     "end": 13
15 },
16 "right_arm_gripper": {
17     "start": 13,
18     "end": 14
19 },
20 "left_arm_eef": {
21     "start": 14,
22     "end": 21
23 },
24 "right_arm_eef": {
25     "start": 21,
26     "end": 28
27 }
28 },
29 "action": {
30     "left_arm_qpos": {
31         "start": 0,
32         "end": 6
33     },
34     "left_arm_gripper": {
35         "start": 6,
36         "end": 7
37     },
38     "right_arm_qpos": {
39         "start": 7,
40         "end": 13
41     },
42     "right_arm_gripper": {
43         "start": 13,
44         "end": 14
45     },
46     "left_arm_eef": {
47         "start": 14,
48         "end": 21
49     },
50     "right_arm_eef": {
51         "start": 21,
```

```
52     "end": 28
53   }
54 },
55 "video": {
56   "cam_high": {
57     "original_key": "observation.images.head_camera"
58   },
59   "cam_left_wrist": {
60     "original_key": "observation.images.left_camera"
61   },
62   "cam_right_wrist": {
63     "original_key": "observation.images.right_camera"
64   },
65 },
66 "annotation": {
67   "human.task_description": {
68     "original_key": "task_index"
69   }
70 }
71 }
```